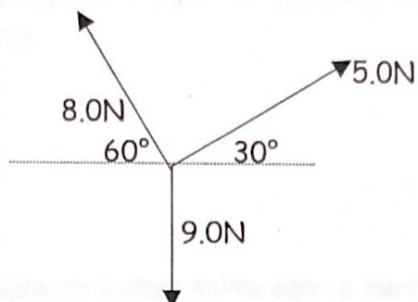


Equilibrium Worksheet#1

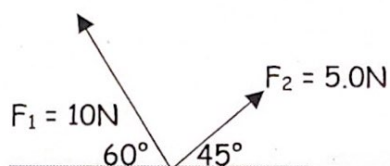
(name) _____

1. What is the magnitude and direction of the force necessary to produce equilibrium in the following?



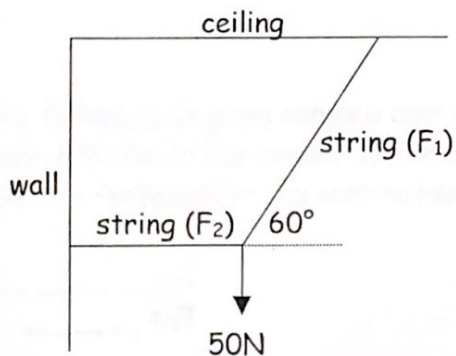
$F =$ _____.

2. What is the magnitude and direction of the force needed to produce equilibrium with F_1 and F_2 ?



$F =$ _____.

3. An object on which the force of gravity is 50 N is supported by two strings as shown below. Calculate the magnitude of the force of tension exerted by each string.



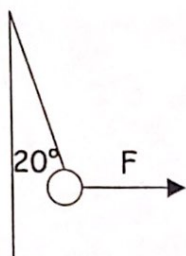
$F_1 =$ _____.

$F_2 =$ _____.

4. A boy weighing 400 N hangs on the middle of a rope stretched between two trees. The rope sags in such a way that it makes an angle of 170° at the boy's hands. What is the tension in each rope?

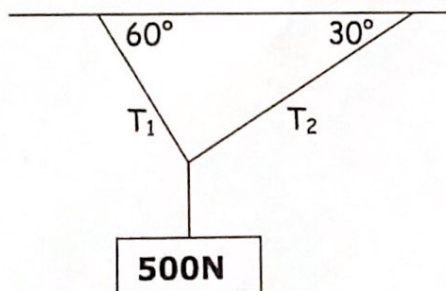
$$T = \underline{\hspace{2cm}}$$

5. A 660 N object is suspended from a rope which makes an angle of 20° with the wall as shown below. What is the magnitude of the horizontal force necessary to produce equilibrium?



$$F = \underline{\hspace{2cm}}$$

6. A 500 N crate is suspended from a cable in such a manner that the cable makes an angle of 30° with the ceiling at one end and an angle of 60° with the ceiling at the other end. Calculate the tension force exerted by each cable on the crate.



$$T_1 = \underline{\hspace{2cm}}$$

$$T_2 = \underline{\hspace{2cm}}$$

Answers: 1. 0.54N, 53° S of W

4. 2.29×10^3 N

5. 240N

2. 12.3N, 83° S of E

3. 29N , 58N

6. $T_1 = 433\text{N}$, $T_2 = 250\text{N}$